

State of the Art: Newton Globalization and Solver Performance for Implicit CFD

Survey for the SFEM CVFEM semi-structured multigrid solver

September 5, 2026

Headline finding

Our solver needs **~ 1000** Krylov iterations per Newton step and fails above $\text{Re} \approx 200$. Published incompressible Navier–Stokes solvers using an augmented-Lagrangian preconditioner need **4–8** iterations per Newton step and reach $\text{Re} = 10\,000$. Plain Newton with *no* globalization at all is documented to reach $\text{Re} \approx 2500$.

Meanwhile our *cost per cycle* is normal: our V-cycle runs at 6.06 M nodes/s, and FUN3D on the same Grace silicon completes a full nonlinear iteration at ≈ 6.0 M points/s.

The conclusion is therefore that the linear solve, and specifically the smoother, is the binding constraint — not the continuation strategy.

1 Scope and baseline

This surveys how established implicit CFD codes get Newton to converge at high Reynolds number, and what throughput they achieve, in order to judge our own numbers. Our reference configuration throughout:

Discretisation	colocated FV/CVFEM, hexahedra, Rhie–Chow stabilisation
Solver	Newton + geometric multigrid V-cycle, block-Jacobi smoother, block size 4
Hardware	one NVIDIA Grace socket, 72 Neoverse V2 cores
Measured	242 500 dof, one V-cycle ≈ 10 ms; 646 MDOF/s operator apply
Converged run	$\text{Re} = 100$, 33 124 dof: 32 Newton steps, $\approx 30\,000$ linear iterations
Failure	diverges above $\text{Re} \approx 200$ –400

2 Pseudo-transient continuation: the formulas

The underlying iteration adds a pseudo-time term to the Jacobian, recovering Newton as $\Delta t \rightarrow \infty$:

$$\left[\frac{M}{\Delta t} + \frac{\partial R}{\partial U} \right] \Delta U^k = -R(U^k), \quad U^{k+1} = U^k + \omega^k \Delta U^k, \quad \text{CFL} = \frac{\lambda_{\max} \Delta t_{\kappa}}{L_{\kappa}}. \quad (1)$$

Convergence theory: Kelley & Keyes, *SINUM* **35**(2), 1998, 508–523, [doi:10.1137/S0036142996304796](https://doi.org/10.1137/S0036142996304796).

2.1 SER — switched evolution relaxation

Mulder & van Leer, *JCP* **59** (1985) 232–246 ([PDF](#)). The CFL is scaled by the inverse residual ratio:

$$\text{CFL}^k = \min \left(\text{CFL}^{k-1} \frac{\|R^{k-1}\|_2}{\|R^k\|_2}, \text{CFL}_{\max} \right) \quad (2)$$

A variant references the *initial* residual instead of the previous one:

$$\text{CFL}^n = \frac{\text{CFL}_0}{r_n}, \quad r_n = \max\left(\frac{\|R(q^n)\|_2}{\|R(q^0)\|_2}, \frac{\|R(q^n)\|_\infty}{\|R(q^0)\|_\infty}\right) \quad (3)$$

PETSc's `TSPSEUDO` implements both forms, with growth factor `-ts_pseudo_increment` and cap `-ts_pseudo_max_dt`.

2.2 EXPur — driven by the under-relaxation factor

Ceze & Fidkowski, AIAA 2013-2686 (PDF), eqn. (13):

$$\text{CFL}^{k+1} = \begin{cases} \beta \text{CFL}^k & \omega^k = 1 \quad (\beta > 1) \\ \text{CFL}^k & \omega_{\min} < \omega^k < 1 \\ \kappa \text{CFL}^k & \omega^k < \omega_{\min} \quad (\kappa < 1) \end{cases} \quad (4)$$

Parameters used: $\omega_{\min} = 0.01$, $\beta = 1.05\text{--}2.0$, $\kappa = 0.1$.

2.3 RDM — residual difference method (bounded SER)

Same reference, eqn. (14). Growth is capped at β , with equality only when the residual vanishes:

$$\text{CFL}^k = \min\left(\text{CFL}^{k-1} \beta^{[(\|R^{k-1}\|_2 - \|R^k\|_2) / \|R^{k-1}\|_2]}, \text{CFL}_{\max}\right) \quad (5)$$

The *monotonic* variant mRDM forces the exponent to zero when the residual increases.

3 What the production codes actually do

3.1 FUN3D: HANIM

Wang, Diskin, Nielsen & Liu, AIAA 2021-0857 (PDF). The legacy scheme was a *linear* CFL ramp specified as two namelist arrays (CFL 1 $\rightarrow$ 500 over 200 iterations) with 15 multicolour Gauss–Seidel sweeps and no Krylov method.

HANIM replaced it with a module hierarchy — preconditioner, GCR, realizability check, nonlinear control, CFL adaption — in which **any module reporting failure discards the update, cuts the CFL, and repeats from the same state**:

success	$\text{CFL} \leftarrow \text{CFL} \times 2$
failure	$\text{CFL} \leftarrow \text{CFL} \times 0.1$
bounds	$\text{CFL}_{\min} = 10^{-12}, \text{CFL}_{\max} = 10^{10} \quad (22 \text{ orders})$

On failure only the *Jacobian diagonal* is updated for the new CFL; fluxes are not re-linearized. The Krylov success criterion is deliberately loose, $\|r_k\|/\|r_0\| \leq \mu_{\text{GCR}} = 0.92$ — an 8 % reduction per outer step. The line search is a *quadratic search with a closed-form minimiser* that reuses the linear residual GCR has already computed, so it costs one extra residual evaluation rather than a sequence:

$$\omega = -\frac{g(1) - f(0)}{2(f(1) - g(1))}, \quad g(1) = \left\| \frac{V}{\Delta\tau} \Delta Q + \frac{\partial R}{\partial Q} \Delta Q + R(Q^n) \right\| \quad (6)$$

Case	Grid	Cores	Iterations (legacy $\rightarrow$ HANIM)	Speedup
Hemisphere-cylinder	9.0 M	160	8976 $\rightarrow$ 675	2.6 $\times$
Hemisphere-cylinder	71.4 M	1400	10 907 $\rightarrow$ 1874	2.1 $\times$
Supersonic duct	3.17 M	240	9721 $\rightarrow$ 1283	2.2 $\times$
Juncture flow	13.0 M	320	no convergence in 72 h $\rightarrow$ converged	6.8 $\times$
Juncture flow	37.1 M	756	—	7.8 $\times$

3.2 NASA LAURA / HyperSolve: the most directly implementable controller

Thompson & O’Connell (NASA PDF), Algorithm 1, verbatim:

```
CFL <- 1.0 ; Q_safe <- Q
while |R(Q)|_2 > tolerance:
  Dt_i <- CFL * V_i / sum_f (|u_f| + c) A_f
  dQ <- (M/Dt + dR/dQ)^-1 R_t
  eta <- line search s.t. |R_t(Q)|_2 > |R_t(Q + eta dQ)|_2
  if eta > 0.1: CFL <- 1.5*CFL ; Q <- Q + eta dQ ; Q_safe <- Q
  else:        CFL <- 0.1*CFL ; Q <- Q_safe
```

The critical detail is that the time term appears on *both* sides, so the merit function and the Jacobian stay consistent: $R_t = R + (M/\Delta t)\Delta Q$ and $[M/\Delta t + \partial R/\partial Q]\Delta Q = -(M/\Delta t)\Delta Q - R(Q^n)$.

3.3 SU2: the shipping controller

From `config.template.cfg`: CFL.ADAPT.PARAM = (0.1, 2.0, 10.0, 1e10, 0.001, 0), i.e. factor-down, factor-up, min, max, *acceptable linear-solver convergence*, start iteration. The fifth parameter matters for us: **SU2 cuts the CFL when the linear solver fails to converge sufficiently**. A naive SER controller ramps on residual history alone and will happily accelerate while the linear solve silently degrades. Guidance: Euler and laminar tolerate doubling to a very high cap; RANS wants factor-up ≈ 1.2 and $\text{CFL}_{\text{max}} \approx 10^3$.

4 Reynolds continuation versus pseudo-transient

The split is by regime, not preference. Compressible aerodynamic codes (FUN3D, SU2, CFL3D, OVERFLOW, LAURA) use PTC exclusively: there is no natural continuation parameter, and pseudo-time is the physically meaningful homotopy. The *finite-element incompressible* community uses Reynolds continuation, because ν is an explicit coefficient and the Stokes problem at $\text{Re} \rightarrow 0$ is trivially solvable and supplies a free starting point.

The reference for our regime

Farrell, Mitchell & Wechsung, *SISC* **41**(5), A3073–A3096 (arXiv:1810.03315). Augmented-Lagrangian preconditioner for 3D stationary incompressible Navier–Stokes at high Reynolds number.

Their schedule: solve **Stokes** first, then $\text{Re} = 10 \rightarrow 100$, then in steps of 100 up to $\text{Re} = 10\,000$, each solution seeding the next. Newton globalized with a PETSc L^2 line search.

4.1 Their iteration counts — the benchmark we should measure against

Krylov iterations per Newton step, 3D lid-driven cavity:

Refinements	dof	Re=10	100	1000	5000	10 000
1	1.0×10^4	2.50	4.33	6.00	8.00	14.00
4	6.6×10^5	2.50	2.67	5.00	8.00	8.50

For contrast, PCD/LSC preconditioning (IFISS) needs 22–26 at $\text{Re}=10$ and 100–210 at $\text{Re}=1000$. Against SIMPLE on the same problem, serial:

Re	AL iters	AL time	SIMPLE iters	SIMPLE time
10	4	0.21 min	515	1.06 min
100	8	0.38	979	2.01
200	10	0.46	1185	2.48

Each SIMPLE iteration is $22\text{--}26\times$ cheaper and it still loses. **Our ~ 1000 cycles per Newton step sits in the SIMPLE column.**

4.2 The alternative that needs no continuation at all

Rebholz et al. ([arXiv:2402.12304](#)) compose one Picard step with one Newton step. The Picard half supplies the stability Newton lacks — global stability for plain Newton does *not* hold — while retaining local quadratic convergence.

Method (same mesh, zero initial guess, no continuation)	Converges to
Newton	$\text{Re} \lesssim 2500$
Newton + line search	$\text{Re} \lesssim 3000$
Picard	$\text{Re} \lesssim 4000$
Picard–Newton	$\text{Re} \approx 10\,000\text{--}12\,000$

Plain Newton’s own documented limit is $\text{Re} \approx 2500$, an order of magnitude above our $\text{Re} \approx 200\text{--}400$. This is the strongest single piece of evidence that our problem is the linear solve rather than the globalization.

5 Inexact Newton: Eisenstat–Walker forcing terms

Eisenstat & Walker, *SISC* **17** (1996) 16–32 ([PDF](#)). This directly attacks over-solving the linear system early in the Newton sequence. PETSc’s default (version 2):

$$\eta_k = \gamma \left(\frac{\|F_k\|}{\|F_{k-1}\|} \right)^\alpha, \quad \alpha = \frac{1+\sqrt{5}}{2} \approx 1.618, \quad \gamma = 1, \quad (7)$$

with $\eta_0 = 0.3$, $\eta_{\max} = 0.9$, and a safeguard $\eta_k \leftarrow \max(\eta_k, \gamma \eta_{k-1}^\alpha)$ when that exceeds 0.1.

6 Smoother strength: local Fourier analysis for Stokes

Claus & Bolten ([arXiv:2008.08719](#)), asymptotic convergence factors:

Smoother	ρ (periodic)	ρ (Dirichlet)
Vanka	0.08	0.10
Triad Gauss–Seidel	0.26	0.36
Triad Jacobi	0.49	<i>divergent</i>

Our point-block Jacobi is weaker than triad-Jacobi, which that study reports degrades to a *divergent* method under Dirichlet boundary conditions — the case we run.

7 Throughput: are our per-cycle costs reasonable?

7.1 FUN3D

Metric	Value	Hardware	Problem
Full nonlinear iteration	≈ 6.0 M points/s	GH200 Grace, 72 cores	5 M points
3000 iterations	2818 s / 1639 s / 430 s	SKL node / EPYC node / A100	3.68 M vertices
Derived	3.91 / 6.73 / 25.6 M points/s	SKL / EPYC / A100	same
15 multicolour GS sweeps	166 ms $\rightarrow$ 30.9 ms	SKL $\rightarrow$ A100 (81 % of peak BW)	1.12 M vertices

The comparison. Our V-cycle is 242 500 dof/10 ms = 6.06 M nodes/s. FUN3D on the same Grace silicon completes an entire nonlinear iteration (residual, Jacobian, 15 Gauss–Seidel sweeps) at ≈ 6.0 M points/s. Our V-cycle costs about what a full production nonlinear iteration costs. NASA’s own framing: “*FUN3D is memory bound: in general, performance scales with memory bandwidth.*”

7.2 OpenFOAM: the throughput-versus-loading curve

Galeazzo et al. 2024, one HLRS Hawk node (2 $\times$ EPYC 7742, 128 cores) (IOP MSE 1312 012009):

Cells/rank	kcells/s/core	L3 miss:hit	MPI ratio	Parallel eff.
210 938	24.4	0.823	0.056	0.96
62 500	39.1	0.520	0.089	0.92
26 367	100.8	0.083	0.279	0.77
15 259	106.0 (peak)	0.054	0.353	0.71
7 813	104.7	0.050	0.469	0.65
3 296	82.0	0.095	0.625	0.54
122	7.0	0.342	0.875	0.19

Two consequences for us. First, our 842 nodes/core sits on the *steep left flank*, between the 3296 and 122 rows, where MPI and parallel overhead dominate — so our measured throughput is more likely **understated** than inflated, and re-measuring at ≈ 15 000 nodes/core (≈ 4.3 M dof on 72 cores) should *raise* it. Second, peak throughput and peak parallel efficiency move in opposite directions: the peak sits at 71 % efficiency, not 95 %. Both numbers should be reported.

Our V-cycle costs 84 knodes/s/core, inside OpenFOAM’s 24–106 kcells/s/core band for a *complete segregated timestep*. Two independent codes therefore agree that our per-cycle cost is production-normal.

8 Ranked recommendations

1. **Picard $\rightarrow$ Newton composition.** We already have both linearizations: Newton’s Jacobian minus the $b^*(u^{k+1}, \hat{u}, v)$ term *is* Picard. Largest robustness gain per line changed; reaches Re ≈ 10 000 with no continuation at all.
2. **Eisenstat–Walker forcing terms.** A few lines; directly attacks over-solving.
3. **Stokes start plus Re continuation** on Farrell’s schedule, replacing our current weak two-stage version.
4. **Full multigrid / nested iteration.** We already own the hierarchy; near-free.
5. **PTC with the LAURA controller** (≈ 30 lines, fully specified above, including the Q_safe rollback that makes it safe).
6. **Per-step physicality limiting** ($\eta_{\max} = 10$ % fractional change).

7. **HANIM in full**, if 1–6 prove insufficient.

Caveat on the ordering

None of items 1–6 will rescue a solver that needs 1000 V-cycles per Newton step. The smoother must be measured alone — as a stationary iteration run to its asymptote, with no coarse space — before any of this is worth implementing.

9 Where comparison is unsound

- FUN3D’s published numbers are **compressible**, block size 5. Its incompressible path (block size 4) has no published performance figures.
- Farrell et al. use a **finite-element** discretisation with an augmented-Lagrangian preconditioner; we use colocated FV with Rhie–Chow. The iteration counts are comparable as a target, the preconditioner is not directly transplantable.
- OpenFOAM figures are per *segregated timestep*, not per Newton step.
- The OHC-1 benchmark paper reports mutually inconsistent absolute MFVOPS figures (§4.2.2 versus Table 3); only its *ratios* should be cited.
- The SU2 smoother details (BCSR 5×5 ; ILU(0) tolerating roughly half the CFL of a GMRES smoother) were **not verified against the primary text** and should be confirmed before being designed around.